The map as one conditional theorem: H1 and H2 give a volume-uniform lattice gap bound, and continuum existence stays a separate obligation

Later corrections: reasons to stop rejects certification of H2 at $\beta_W \simeq 6$ by any certified method known here; the bands note (correction of 2026-10-02) makes box sides 3–5 trial sizes, withdraws the reduction to three open blocking steps and corrects the energy conversion; the position note (correction of 2026-10-02) fixes the strong-coupling threshold set; the October 1 audit §3 lists this note's overclaims (relevant passages read).

Correction (2026-10-02). A referee report and the October 1 audit found that the conclusion outran the hypotheses. The changes, made in place below:

1. *Scope.* "Nothing else is missing", "a proof of the mass gap for $SU(3)$ consists of H1 and H2", "the repository's formal endpoint" and "progress ... on H1 or H2, and nothing else" are **withdrawn**. H1 and H2 yield lattice statements: decay of correlations and a volume-uniform gap of the Wilson transfer matrix at each spacing of the trajectory, and the limit of that bound. Convergence of Schwinger functions, the Osterwalder–Schrader axioms, the infinite-volume continuum theory, nontriviality and T3 are separate obligations. The former item 4 is split into a lattice statement (item 4) and a continuum statement conditional on a separately constructed limit (item 5, hypothesis (E)).
2. *Prefactors.* The constants C_k may grow with $K(a_0)$. The proof now shows why that is harmless: at fixed a_0 the spectral theorem turns decay with any finite prefactor into a gap whose prefactor is $\|\psi\|^2$, and that prefactor survives a continuum limit.
3. *H1.* The former H1 bounded fluctuation covariances only for observables local on the k-th lattice. The proof applied it to conditional expectations of fine observables, which are only quasi-local there, and its coarse remainder needed a locality property that was never assumed. H1 now states (a) the fluctuation bound for those projections, (b) quasi-locality of the coarse projection, and (c) a gauge-invariant Gibbs representation on periodic boxes. The cross terms of the proof vanish exactly (Lemma 2), replacing "bounded by the same rates".
4. *H2 and periodic boxes.* Strong mixing of Φ_0 on one box with a robustness margin gives no decay in other volumes without a further theorem. H2 now assumes strong mixing for every gauge-invariant interaction of the r-ball on every coarse periodic box of side at least L_0, and Lemma 1 derives decay there. The reduction to one finite box is a finite-size criterion (Dobrushin–Shlosman; Martinelli–Olivieri) whose transfer to link variables with $SU(3)$ spins, exponentially decaying interactions and periodic boxes is **assumed, not derived**. The reading "on the gauge-invariant content of W, the small Wilson loops" is replaced by an exact statement, Lemma 0, whose invariant σ-algebra includes boundary-attached open lines. Box sides 3–5 are trial sizes, and certification at $\beta_W \simeq 6$ is out of reach of every certified method known here. Item 1 no longer claims a unique infinite-volume Gibbs state.
5. *Numbers.* The glueball energy, printed as $1.66\,\hbar c\,\mathrm{fm}^{-1}$, is $8.42\,\hbar c\,\mathrm{fm}^{-1} \simeq 1.66$ GeV, the value behind $\mu_{0^{++}} a \simeq 0.78$. The spacing a_* is 0.093 fm at $\beta_W = 6.0$ and 0.17 fm at $\beta_W = 5.7$ with $r_0 = 0.5$ fm; the former lead's "0.1 to 0.17 fm" rounds the first figure and §4's 0.093 fm is the same endpoint. These spacings and the rate 0.78 are continuum-scaled estimates, and identifying a_* with a published spacing is heuristic.

6. *What is in place.* "Both hypotheses are finite in kind" and H1's "non-perturbative part is three steps" are **withdrawn**, following the bands note. "T1, T2 ... all in place with explicit constants" is corrected: the volume-uniform Kogut–Susskind gap rests on Yarotsky's existential threshold, and the explicit rigorous thresholds belong to the Wilson transfer matrix, $g^2 > 444$ (Dobrushin) and $g^2 \geq 1059$ (polymer expansion, all representations), with 176 the leading-activity figure only. The theorem uses none of them.

The programme's pure-$SU(3)$ map can be stated as one conditional theorem about the Wilson lattice theory. Hypothesis H1 controls the blocking steps from the bare spacing a_0 up to a spacing a_*, where the effective interaction lies within a radius r of a reference interaction Φ_0. Hypothesis H2 asks every gauge-invariant interaction of that neighbourhood to be strongly mixing, uniformly over periodic boxes. The conclusion is a volume-uniform lower bound on the gap of the Wilson transfer matrix at each spacing of the trajectory,

$$\Delta(a_0) \;\geq\; \frac{\hbar c}{a_0} \cdot \frac{\min(\gamma_*, \gamma')}{2^{K(a_0)}}, \qquad \liminf_{a_0 \to 0} \Delta(a_0) \;\geq\; \frac{\hbar c \, \min(\gamma_*, \gamma')}{a_*},$$

where γ_* is the decay rate of H1, γ' the mixing rate of H2, both per coarse lattice unit, and $2^{K(a_0)} a_0 \to a_*$. If Φ_0 is the Wilson interaction at β_W between 5.7 and 6.0, then a_* is, heuristically, the published spacing there: from 0.17 fm down to 0.093 fm with $r_0 = 0.5$ fm.

H1 and H2 leave the continuum construction open. They give no convergence of Schwinger functions as $a_0 \to 0$, no Osterwalder–Schrader axioms (Euclidean invariance in particular), no infinite-volume continuum theory and no nontriviality. The same number bounds the gap of a continuum Hamiltonian only for a continuum limit constructed separately, with the axioms, in the sense of hypothesis (E) of §3. The upper side $m < \infty$ needs nontriviality, and T3 is not implied. Neither hypothesis is available today: no blocking step of H1 is controlled with explicit constants, and H2 has no feasible certified route at $\beta_W \simeq 6$. Constants explicit; nothing promoted.

1. Setting

Measures. The Wilson measure μ_{β_0} lives on the links of a periodic box in $(a_0\mathbb{Z})^4$ at bare coupling $\beta_0 = 6/g_0^2$. A block-spin map $\mathcal{B}$ of block size 2 is a gauge-covariant assignment of a coarse link variable to each link of the coarse lattice, measurable in the fine links, such that coarse Wilson loops are functions of fine Wilson loops. The renormalized measures $\mu^{(k)} = \mathcal{B}^k \mu_{\beta_0}$ live on the links of the periodic coarse lattices of spacing $2^k a_0$ and reproduce exactly the correlations of coarse observables. Box sides are multiples of $2^K a_0$, so that every level tiles.

Conditional expectations. Let $\mathcal{G}_k$ be the σ-algebra generated by the level-k links. Then $\mathcal{G}_0$ is generated by all fine links, and $\mathcal{G}_{k+1} \subseteq \mathcal{G}_k$ because level-$(k+1)$ links are functions of level-k links. Write $E_k = \mathbb{E}_{\mu_{\beta_0}}[\,\cdot \,|\, \mathcal{G}_k]$ and $D_k = E_k - E_{k+1}$. For a function F of level-k links, $E_{k+1}F$ is the conditional expectation given the coarse links, written $\bar{F}$ in the earlier version.

Distances and supports. Distances are ℓ^∞ distances between link midpoints, measured on the torus in units of the lattice concerned. A local observable is a bounded measurable function of the links in a finite set, its support S_F. For a set S of fine links, $S^{(K)}$ is the set of level-K links within level-K distance $c_{\mathcal{B}}$ of the rescaled set $2^{-K}S$, where $c_{\mathcal{B}}$ is a constant of the block map. Fine sets at fine distance d therefore have images at level-K distance at least $d/2^K - 2c_{\mathcal{B}}$.

Interactions. An interaction $\Phi = (\Phi_X)$ assigns a bounded function of the links of X to each finite set X of links; it is gauge-invariant if every Φ_X is. Its conditional Gibbs measure on a set Λ of links, given a configuration ω outside Λ, is

$$\mu_\Lambda^{\Phi,\omega}(dU) \;\propto\; \exp\Big[- \sum_{X \cap \Lambda \neq \emptyset} \Phi_X(U \vee \omega) \Big] \prod_{\ell \in \Lambda} dU_\ell,$$

with Haar measure dU_ℓ. With range weight $\kappa > 0$ the interaction norm is $\|\Phi\|_\kappa = \sup_\ell \sum_{X \ni \ell} e^{\kappa \operatorname{diam} X} \|\Phi_X\|_\infty$, using the torus diameter on a periodic box. Total variation is $\|\mu - \nu\|_{\mathrm{TV}} = \sup_A |\mu(A) - \nu(A)|$, so that $|\mu(f) - \nu(f)| \le 2\|f\|_\infty \|\mu - \nu\|_{\mathrm{TV}}$ for bounded f.

2. The hypotheses

H2 (uniform strong mixing near Φ_0). There are a gauge-invariant reference interaction Φ_0 of finite range on the coarse links, numbers $\kappa > 0$, $r > 0$, $C' < \infty$, $\gamma' > 0$ and an integer L_0 with the following property. Let T be a coarse periodic box whose sides are at least L_0, and Φ a gauge-invariant interaction on T with $\|\Phi - \Phi_0\|_\kappa \le r$ (Φ_0 periodized). Then for every set Λ of links of T, every $W \subseteq \Lambda$, and every pair of configurations ω, ω' on $T \setminus \Lambda$ that differ only at one link y,

$$\big\| \mu_\Lambda^{\Phi,\omega}\big|_W - \mu_\Lambda^{\Phi,\omega'}\big|_W \big\|_{\mathrm{TV}} \;\le\; C'\,|W|\, e^{-\gamma'\, d(W,y)}.$$

The proof below uses this only for Λ the complement in T of a finite set.

H1 (the weak side). Along a family of bare couplings with spacings $a_0 \to 0$ (the trajectory) there are integers $K = K(a_0)$ with $2^K a_0 \to a_*$ and a number $\gamma_* > 0$, with the following properties on every fine periodic box whose sides are multiples of $2^K a_0$ and at least $2^K L_0 a_0$ (Φ_0, κ, r, L_0 as in H2). The constants $C_k(F,G)$ and $B(F)$ may depend on a_0, k and the observables, but not on the box.

- (a) *Fluctuations.* For $k < K$ and local fine F, G at fine distance d,

$$\big|\mathbb{E}_{\mu_{\beta_0}}\big[(D_k F)(D_k G)\big]\big| \;\le\; C_k(F,G)\, e^{-\gamma_* d/2^k}.$$

- (b) *Quasi-locality of the coarse projection.* For local fine F and every integer $\ell \ge 0$ there is a function F_ℓ of the level-K links within level-K distance ℓ of $S_F^{(K)}$ with $\|E_K F - F_\ell\|_\infty \le B(F)\, e^{-\gamma_* \ell}$.

- (c) *Gibbs representation.* $\mu^{(K)}$ has density proportional to $\exp[-\sum_X \Phi_X^{(K)}]$ with respect to Haar measure on the coarse box, for a gauge-invariant interaction $\Phi^{(K)}$ with $\|\Phi^{(K)} - \Phi_0\|_\kappa \le r$.

For F, G functions of level-k links, the left side of (a) equals $\langle F; G\rangle_{\mu^{(k)}} - \langle \bar F; \bar G\rangle_{\mu^{(k+1)}}$, the quantity of the earlier H1. That version stopped at such F, G; the proof applies (a) to $E_k F$ for fine F, which is a function of level-k links but only quasi-local on that lattice. In words: the fluctuation added at each step clusters exponentially at the scale of that step, the coarse projection of a local observable is exponentially well localized at the final scale, and the trajectory arrives within distance r of Φ_0.

What is known about H1. At small effective coupling, rigorous small-field renormalization (Balaban and others, metadata level, as listed in reasons to stop) controls effective actions in finite volume with unspecified constants; no source read here derives (a)–(c) from it. For the repository's one explicit small-field step, a scaling estimate with crude constants puts the remainder threshold at $g^2 \sim 10^{-12}$ if an unproved order-one propagator decay rate holds, and at $g^2 \sim 10^{-35}$ at the proved rate (position note). The final steps, in which the continuum-scaled ξ/a passes from about 10 to about 1, are the confinement-scale problem; $\log_2 10 \simeq 3.3$ doublings is a kinematic count and does not show that only these steps are open (bands note §§1, 2b). Part (c) can fail for block maps in general: van Enter, Fernández and Sokal (J. Stat. Phys. 72 (1993) 879; abstract read, as recorded in the bands note §2b) give non-Gibbsian images.

Lemma 0 (gauge reduction). Let Φ be gauge-invariant, Λ a set of links, and G_Λ the group of gauge transformations at the sites all of whose incident links lie in Λ. For configurations ω, ω' outside Λ and $W \subseteq \Lambda$, the total-variation distance between the marginals of $\mu_\Lambda^{\Phi,\omega}$ and $\mu_\Lambda^{\Phi,\omega'}$ on W equals the supremum of $|\mu_\Lambda^{\Phi,\omega}(A) - \mu_\Lambda^{\Phi,\omega'}(A)|$ over G_Λ-invariant events A of the links of W.

Proof. An element of G_Λ moves only links of Λ, leaves ω fixed, preserves Haar measure and every Φ_X, and so preserves $\mu_\Lambda^{\Phi,\omega}$; it maps configurations of W to configurations of W. For $|f| \le 1$ on W, the Haar average Pf of f over G_Λ is invariant, satisfies $|Pf| \le 1$, and has the same expectation as f under both measures. Since the total-variation distance is half the supremum of $|\mu(f) - \nu(f)|$ over $|f| \le 1$, the supremum may be taken over invariant f. □

The invariant σ-algebra contains the traces of closed loops in W and the matrix entries of open paths in W whose endpoints are sites with an incident link outside Λ. A selection of small Wilson loops is not known to generate it. So H2 is to be read on this whole σ-algebra, boundary-attached lines included; interior single-link marginals are Haar and contribute nothing (bands note).

From one box to H2. The earlier H2 asked for the mixing bound for Φ_0 on one box V of side R, with a margin covering the r-ball. Statements of that kind are the input of finite-size criteria: the Dobrushin–Shlosman constructive criterion and complete analyticity (*Statistical Physics and Dynamical Systems*, Birkhäuser 1985, 347 and 371; J. Stat. Phys. 46 (1987) 983) and the strong-mixing theory of Martinelli and Olivieri (Commun. Math. Phys. 161 (1994) 447 and 487), all at metadata level. They pass from a quantitative condition on one finite box to mixing in every volume of a class. The proof needs H2 as stated, in every periodic box. Applying those criteria here requires their transfer to link variables with $SU(3)$ spins, to interactions with exponential tails such as $\Phi^{(K)}$, and to periodic boxes; this note assumes that transfer and does not derive it. The robustness margin itself is elementary: replacing Φ_0 by Φ with $\|\Phi - \Phi_0\|_\kappa \le r$ changes the energy of a box V by at most $|V|\, r$ in sup norm, hence each conditional measure on V by at most $e^{2|V|r} - 1$ in total variation (Dobrushin note, Lemma 1).

The reference interaction. A natural Φ_0 is the Wilson interaction at β_W between 5.7 and 6.0, where the continuum-scaled ξ/a runs from 0.70 to 1.28 (bands note, table). A nearby interaction with a small negative adjoint term moves Φ_0 away from the endpoint of the fundamental–adjoint first-order line, where the scalar channel softens (precision of 2026-09-23, openings note §1). Boxes of side $R = 3$ to 5 are trial sizes, and no sufficient size is known. For them a certified bound is an integral over 1728 to 16000 real dimensions with a supremum over boundary conditions (bands note §2). Reasons to stop records that no certified method known here reaches $\beta_W \simeq 6$; certified polymer enumeration is proved only for $\beta_W \lesssim 10^{-2}$ (position note, correction of 2026-10-02).

3. The theorem

Write $\gamma_m = \min(\gamma_*, \gamma')$. A periodic box is admissible for a_0 if its sides are multiples of $2^{K(a_0)}a_0$ and at least $2^{K(a_0)}L_0 a_0$.

Theorem (conditional). Assume H1 and H2. For every a_0 on the trajectory:

1. *Coarse decay.* On the coarse lattice of every admissible box, local observables X, Y at level-K distance $D \ge 1$ satisfy
$$|\langle X; Y\rangle_{\mu^{(K)}}| \;\le\; 2C'\,|S_X|\,|S_Y|\,\|X\|_\infty \|Y\|_\infty\, e^{-\gamma' D}.$$
2. *Fine decay.* On every admissible box, local fine observables F, G at fine distance d satisfy, for every $\varepsilon \in (0, \gamma_m)$,
$$|\langle F; G\rangle_{\mu_{\beta_0}}| \;\le\; A_\varepsilon(F,G)\, e^{-(\gamma_m - \varepsilon)\, d/2^K},$$
with $A_\varepsilon(F,G)$ finite and independent of the box.
3. *Lattice gap.* For every periodic spatial box whose side is a multiple of $2^K a_0$ and at least $2^K L_0 a_0$, the Wilson transfer matrix $\mathcal{T} = e^{-a_0 H_W/\hbar c}$ satisfies, on the gauge-invariant vectors orthogonal to its vacuum,
$$\Delta(a_0) \;\ge\; \frac{\hbar c}{a_0} \cdot \frac{\gamma_m}{2^{K(a_0)}}.$$

4. *Along the trajectory.* $\liminf_{a_0\to 0}\Delta(a_0)\geq \hbar c\,\gamma_m/a_*$, uniformly over the boxes of item 3.
5. *Continuum, conditionally.* If (E) below also holds, the continuum Hamiltonian satisfies $H\geq \hbar c\,\gamma_m/a_*$ on the vectors orthogonal to the vacuum.

(E) Continuum existence along the trajectory. There are a sequence of pairs (a_0, spatial box as in item 3) with $a_0\to 0$, a continuum theory satisfying the Osterwalder–Schrader axioms with vacuum Ω and Hamiltonian $H\geq 0$, and a family $\mathcal{F}$ of gauge-invariant observables supported at positive Euclidean times, each F with lattice approximants F_{a_0}, such that: with ψ_{F,a_0} the lattice vector of F_{a_0} minus its vacuum component and $n=\lfloor\tau/a_0\rfloor$, the inner products $\langle\psi_{F,a_0},\mathcal{T}^n\psi_{F,a_0}\rangle$ converge for every $\tau\geq 0$ to $\langle\psi_F,e^{-\tau H/\hbar c}\psi_F\rangle$; and the vectors ψ_F, $F\in\mathcal{F}$, are dense in $\Omega^\perp$.

H1 and H2 alone give items 1–4. Item 5 is the only continuum statement, and its hypothesis (E) belongs to the construction problem T4 of the obligations map.

Lemma 1 (decay on a periodic box). Let Φ satisfy the bound of H2 on a coarse periodic box T, and let μ be its Gibbs measure on T. Local X,Y at distance $D\geq 1$ satisfy $|\langle X;Y\rangle_\mu|\leq 2C'|S_X||S_Y|\,\|X\|_\infty\|Y\|_\infty e^{-\gamma' D}$.

Proof. Put $\Lambda=T\setminus S_X$. Given the configuration ω on S_X, the links of Λ have law $\mu_\Lambda^{\Phi,\omega}$, the conditional property of a finite-volume Gibbs measure. Hence $\langle X;Y\rangle=\mathbb{E}_\mu\big[X(\mu_\Lambda^{\Phi,\omega}(Y)-\mathbb{E}_\mu Y)\big]$, and since $\mathbb{E}_\mu Y$ is an average of $\mu_\Lambda^{\Phi,\omega'}(Y)$ over ω', $|\langle X;Y\rangle|\leq\|X\|_\infty\sup_{\omega,\omega'}|\mu_\Lambda^{\Phi,\omega}(Y)-\mu_\Lambda^{\Phi,\omega'}(Y)|$. Change ω into ω' one link of S_X at a time. Each step changes the boundary condition at one link $y\in S_X$, at distance at least D from $W=S_Y\subseteq\Lambda$, and by H2 moves $\mu_\Lambda^\Phi(Y)$ by at most $2\|Y\|_\infty C'|S_Y|e^{-\gamma' D}$. There are $|S_X|$ steps. $\square$

Lemma 2 (exact decomposition). For bounded F,G,

$$\mathrm{Cov}_{\mu_{\beta_0}}(F,G)=\mathrm{Cov}(E_KF,E_KG)+\sum_{k=0}^{K-1}\mathbb{E}\big[(D_kF)(D_kG)\big].$$

Proof. Telescoping gives $F=E_KF+\sum_{k<K}D_kF$, since $E_0F=F$. For $j>k$, D_jG is $\mathcal{G}_{k+1}$-measurable and $\mathbb{E}[D_kF\,|\,\mathcal{G}_{k+1}]=0$, so $\mathbb{E}[(D_kF)(D_jG)]=0$; likewise $\mathbb{E}[(D_kF)\,E_KG]=0$ and $\mathbb{E}[D_kF]=0$. Expand $\mathbb{E}[FG]$ and use $\mathbb{E}F=\mathbb{E}\,E_KF$. $\square$

The earlier proof bounded the cross terms by the decay rates; they vanish identically.

Lemma 3 (the coarse term). Assume H1(b), H1(c) and H2. Let local fine F,G have images $S_F^{(K)},S_G^{(K)}$ at level-K distance $D\geq 1$, and put $b_F=(1+e^{\gamma_*})(\|F\|_\infty+B(F))$. Then

$$|\,\mathrm{Cov}(E_KF,E_KG)|\;\leq\;b_Fb_G\Big[32\,C'\,|S_F^{(K)}|\,|S_G^{(K)}|\,(2D+1)^{10}+\frac{D+1}{(1-e^{-\gamma_*})^2}\Big]e^{-\gamma_m D}.$$

Proof. Let $X_i=F_i$ from H1(b) for $i<N$ and $X_N=E_KF$, with N larger than the diameter of the coarse box; put $\Delta X_0=X_0$ and $\Delta X_i=X_i-X_{i-1}$. Then $E_KF=\sum_{i\leq N}\Delta X_i$, $\|\Delta X_i\|_\infty\leq b_Fe^{-\gamma_* i}$, and ΔX_i depends on the set A_i of level-K links within distance i of $S_F^{(K)}$. A point has at most $4(2i+1)^4$ link midpoints within ℓ^∞ distance i, so $|A_i|\leq 4(2i+1)^4|S_F^{(K)}|$. Define ΔY_j and B_j likewise. Then $\mathrm{Cov}(E_KF,E_KG)=\sum_{i,j}\mathrm{Cov}(\Delta X_i,\Delta Y_j)$. For $i+j<D$ the sets A_i,B_j are at distance at least $D-i-j\geq 1$. By H1(c) and H2, Lemma 1 applies to $\mu^{(K)}$, and the term is at most $2C'\cdot 16(2i+1)^4(2j+1)^4|S_F^{(K)}||S_G^{(K)}|\,b_Fb_G\,e^{-\gamma_*(i+j)-\gamma'(D-i-j)}$. Since $\gamma_*(i+j)+\gamma'(D-i-j)\geq\gamma_mD$, and since there are at most $(2D+1)^2$ such pairs, each with $i,j<D$, these terms sum to at most the first bracket. For $i+j\geq D$ use $|\,\mathrm{Cov}(U,V)|\leq\|U\|_\infty\|V\|_\infty$: the $s+1$ pairs with $i+j=s$ give at most $b_Fb_G\sum_{s\geq D}(s+1)e^{-\gamma_*s}\leq b_Fb_G(D+1)e^{-\gamma_*D}/(1-e^{-\gamma_*})^2$. $\square$

Proof of the theorem. *Item 1* is Lemma 1 applied to $\Phi^{(K)}$, which H1(c) places in the r-ball of H2.

Item 2. By Lemma 2 the covariance is the coarse term plus the fluctuation terms. By H1(a), and since $2^k\leq 2^{K-1}$ for $k<K$, the fluctuation terms total at most $\big(\sum_{k<K}C_k(F,G)\big)e^{-2\gamma_*d/2^K}$.

The images are at level-K distance $D \geq d/2^K - 2c_{\mathcal{B}}$. For $D \geq 1$, Lemma 3 bounds the coarse term by a polynomial in D times $e^{-\gamma_m D}$, which is at most $c_\varepsilon(F,G)\, e^{-(\gamma_m-\varepsilon)D}$ with $c_\varepsilon(F,G)$ finite; for $D < 1$ use $|\operatorname{Cov}| \leq \|F\|_\infty \|G\|_\infty$. Both bounds become $e^{-(\gamma_m-\varepsilon)d/2^K}$ times a finite constant. The constant $A_\varepsilon(F,G)$ so obtained may grow with $K(a_0)$; it does not depend on the box.

Item 3. Fix an admissible spatial box and normalize $\mathcal{T}$ so that its top eigenvalue on gauge-invariant vectors is 1, with eigenvector Ω. $\mathcal{T}$ is self-adjoint and positive (Lüscher, Commun. Math. Phys. 54 (1977) 283; Osterwalder and Seiler, Ann. Phys. 110 (1978) 440; metadata level, as in the Wilson note §3). On gauge-invariant functions its kernel is continuous and strictly positive on a compact configuration space, so by Jentzsch's theorem the top eigenvalue is simple and isolated and $\Omega > 0$. For a bounded gauge-invariant F of the spatial links of one time slice put $\psi = (F - \langle \Omega, F\Omega\rangle)\Omega$. On the periodic box of time extent L_t, the truncated Euclidean correlation $\langle F^*; \theta_n F\rangle_{L_t}$ converges to $\langle \psi, \mathcal{T}^n \psi\rangle$ as $L_t \to \infty$ through admissible values. Item 2, with the supports at fine distance n, bounds it by $A_\varepsilon e^{-(\gamma_m-\varepsilon)n/2^K}$ uniformly in L_t. If the spectral measure of ψ gave mass $c > 0$ to $(e^{-\lambda}, 1]$ with $\lambda < (\gamma_m - \varepsilon)/2^K$, then $\langle \psi, \mathcal{T}^n \psi\rangle \geq c\, e^{-\lambda n}$ would contradict that bound for large n. Since ε is arbitrary, the spectral measure of every such ψ lives on $[0, e^{-\gamma_m/2^K}]$. These vectors are dense in the gauge-invariant vectors orthogonal to Ω, because $\Omega > 0$ almost everywhere. Hence $\|\mathcal{T}|_{\Omega^\perp}\| \leq e^{-\gamma_m/2^K}$, and item 3 follows with $\Delta = -(\hbar c/a_0) \log \|\mathcal{T}|_{\Omega^\perp}\|$. The constant A_ε entered only through its finiteness.

Item 4. $\hbar c\, \gamma_m/(2^{K(a_0)} a_0) \to \hbar c\, \gamma_m/a_*$ by H1.

Item 5. For $F \in \mathcal{F}$ the vector ψ_{F,a_0} is orthogonal to the lattice vacuum, so item 3 and the positivity of $\mathcal{T}$ give $\langle \psi_{F,a_0}, \mathcal{T}^n \psi_{F,a_0}\rangle \leq \|\psi_{F,a_0}\|^2 e^{-n a_0 \Delta(a_0)/\hbar c}$, and $n a_0 \Delta(a_0)/\hbar c \geq \lfloor \tau/a_0 \rfloor a_0 \gamma_m/(2^K a_0) \to \tau \gamma_m/a_*$. The case $\tau = 0$ of (E) gives $\|\psi_{F,a_0}\|^2 \to \|\psi_F\|^2$. Hence $\langle \psi_F, e^{-\tau H/\hbar c} \psi_F\rangle \leq \|\psi_F\|^2 e^{-\tau\gamma_m/a_*}$ for all $\tau \geq 0$. The spectral measure of H in ψ_F therefore lives on $[\hbar c \gamma_m/a_*, \infty)$, by the argument of item 3, and density of the ψ_F gives the claim. □

The constants C_k, $B(F)$ and A_ε, which may grow with $K(a_0)$, never reach item 5: the passage to the continuum needs (E) and no uniformity in those constants.

Jaffe–Witten and T3. Item 5 supplies the lower half of the Jaffe–Witten clause $0 < m < \infty$ for a continuum theory constructed by other means. The upper half is the bound $m \leq M_0/M_{-1}$ of the moment-hierarchy note, which assumes the Osterwalder–Schrader axioms and a flowed correlator $C_0 \not\equiv 0$, that is, nontriviality. T3 asks the scaled gap $\Delta(a_0)/(\hbar c \Lambda)$ to converge. Item 4 bounds its lower limit by $\gamma_m/(a_* \Lambda)$ and is silent on convergence. The finite-verification note §4 relates T3 to the convergence of blocking trajectories under its assumption of exact low-energy reductions; that relation is heuristic.

4. What the theorem makes visible

The bound is a ratio of a rate to a physical length. Suppose Φ_0 is the Wilson interaction at $\beta_W = 6.0$ and $\Phi^{(K)}$ lies close to it. Heuristically a_* is then the published spacing at that coupling, $a_* \simeq 0.093$ fm ($r_0/a = 5.37$, $r_0 = 0.5$ fm). The continuum-scaled glueball estimate $E_{0^{++}} = 8.42\, \hbar c\, \mathrm{fm}^{-1} \simeq 1.66$ GeV gives $\mu_{0^{++}} a_* \simeq 0.78$ (bands note, table; a continuum-scaled value, not a measured lattice mass). A proved γ_m of any positive size gives item 4. A rate $\gamma_m = 0.1$ would give $\hbar c\, \gamma_m/a_* \simeq 1.07\, \hbar c\, \mathrm{fm}^{-1} \simeq 0.21$ GeV, about one eighth of that estimate. At $\beta_W = 5.7$ the same figures are $a_* \simeq 0.17$ fm and $\mu_{0^{++}} a \simeq 1.44$.

What the hypotheses cost. H1 has $K(a_0) \simeq \log_2(a_*/a_0)$ steps, unbounded as $a_0 \to 0$, so it is an infinite family of statements with order-one coupling only at its end; no step is controlled here with explicit constants (§2). H2 is uniform strong mixing in every periodic box for a whole neighbourhood of interactions. It reduces to one finite box only through the assumed transfer of a finite-size criterion, and the certification of that box at $\beta_W \simeq 6$ is out of reach of the certified methods known here.

What is in place, and what is left. In place: T1 for every finite lattice (obligations map §2); the decay-to-gap transfer (item 3; the Wilson note §3); volume-uniform strong-coupling gaps for the Wilson transfer matrix at $g^2 > 444$ (Dobrushin) and $g^2 \geq 1059$ (polymer expansion, all representations), with 176 the leading-activity figure only; T2 for the Kogut–Susskind Hamiltonian from Yarotsky's existential threshold, which his proof makes $g_0^2 \simeq 10^{101}$; and the upper side of the gap given existence and nontriviality. The theorem uses none of the strong-coupling results, since its trajectory comes from weak coupling. Left: H1 and H2, and outside the theorem the construction (E) with the axioms and the infinite-volume limit, nontriviality, and T3.

5. Consequence for STATE

The pure-gauge map is a conditional lattice theorem. H1 and H2 give a volume-uniform gap of the Wilson transfer matrix at each spacing of the trajectory, at least $\hbar c \min(\gamma_*, \gamma')/(2^K a_0)$, whose lower limit is at least $\hbar c \min(\gamma_*, \gamma')/a_*$. In a continuum limit constructed separately with the Osterwalder–Schrader axioms, the same number bounds the Hamiltonian's gap from below. H2 now means uniform strong mixing in every periodic box; it reduces to one box only through an assumed transfer of a finite-size criterion, and it has no feasible certified route at $\beta_W \simeq 6$. The STATE line, with H1/H2 open and continuum construction and nontriviality explicit obligations, stands as written; the catalogs need this note's new title and scope.